

# Corrections and Clarifications

FFGFT / T0 Time-Mass Duality

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## Preliminary Note

This document lists corrections and refinements for previously published documents in the FFGFT series. Older documents are *not* modified. Where a contradiction exists between an older and a newer document, the version recorded here is binding.

Each entry contains: affected document and location, the erroneous expression, the correct expression, and a brief justification.

### 1 Correction 1: Prefactor in the Higgs derivation of $\xi$

#### Affected Document

Doc. 049 (T0 Lagrangian and Higgs Integration), HTML draft version and earlier working versions.

#### Erroneous Expression

$$\xi = \frac{\lambda_h^2 v^2}{64\pi^4 m_h^2} \approx 1.0 \times 10^{-5} \quad (\text{old - wrong})$$

#### Correct Expression

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.33 \times 10^{-4} \quad (190.1)$$

with the experimental inputs:

$$m_h = 125.25 \text{ GeV} \quad (\text{Higgs mass}),$$

$$v = 246.22 \text{ GeV} \quad (\text{vacuum expectation value}),$$

$$\lambda_h = \frac{m_h^2}{2v^2} \approx 0.129 \quad (\text{Higgs self-coupling}).$$

#### Justification

The prefactor  $64\pi^4$  originated as a typo in the early HTML visualization. It yields  $\xi \approx 10^{-5}$ , which deviates by an order of magnitude from the geometrically derived value

$\xi = 4/3 \cdot 10^{-4}$  and does not reproduce the natural constants. The correct prefactor  $16\pi^3$  yields  $\xi \approx 1.33 \times 10^{-4}$ , consistent with the value independently determined from the proton-electron mass ratio (Doc. 009).

## 2 Correction 2: Tau-Yukawa prefactor $r_\tau$

### Affected Document

Doc. 116 (Koide formula as a consequence of T0 theory), table of Yukawa coefficients.

### Erroneous Expression

$$r_\tau = \frac{8}{3} \approx 2.667 \Rightarrow m_\tau \approx 1712 \text{ MeV} \quad (-3.6\% \text{ dev.})$$

(old - wrong)

### Correct Expression

$$\boxed{r_\tau = \frac{25}{9} \approx 2.778} \Rightarrow m_\tau \approx 1783 \text{ MeV} \quad (+0.4\% \text{ dev.})$$

(190.2)

The lepton masses are generally given by:

$$m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v \quad (190.3)$$

with exponents  $p_e = 3/2$ ,  $p_\mu = 1$ ,  $p_\tau = 2/3$  (step size  $\Delta p = 1/3$ ) and the vacuum expectation value  $v$ .

### Justification

$r_\tau = 8/3$  was a remnant from an earlier iteration stage. Doc. 158 (Koide-to-g-2 bridge) already uses  $r_\tau = 25/9$ , which agrees numerically with the tau mass. Furthermore,  $25/9 = (5/3)^2$  has a geometric interpretation as the square of the pure sixth in the natural overtone series, consistent with the musical resonance analogy (Doc. 060).

Complete Corrected Prefactor Table

| Lepton   | Exponent <i>p</i> | Prefactor <i>r</i> | <i>m</i> [MeV] |
|----------|-------------------|--------------------|----------------|
| Electron | 3/2               | 4/3                | 0.511          |
| Muon     | 1                 | 16/5               | 105.7          |
| Tau      | 2/3               | 25/9               | 1783           |

3 Correction 3: Base formula for *g*-2 of the electron

Affected Document

T0 Explorer (HTML visualization), first version; earlier draft versions of Doc. 018.

Erroneous Expression

$$a_e = \frac{4\pi}{f \cdot k_{\text{geom}}}$$

(old – wrong)

Correct Expression

$$a_e = \frac{S_3}{f \cdot k_{\text{geom}}} = \frac{2\pi^2}{f \cdot k_{\text{geom}}}$$

(190.4)

with  $f = 1/\xi \approx 7500$ ,  $k_{\text{geom}} = (2/\sqrt{\varphi}) \cdot \sqrt{2} \approx 2.224$  and  $S_3 = 2\pi^2 \approx 19.74$  (surface area of the 4D unit sphere, i.e. the 3D boundary of the torus).

Justification

The earlier prefactor  $4\pi$  corresponds to the surface area of the 2D sphere in 3D space, not the geometrically correct 3D surface of the 4D torus.  $S_3 = 2\pi^2$  is the correct value from torus geometry. The fractal corrections for muon and tau still use  $4\pi$ :

$$\Delta a_\mu = \frac{4\pi}{f^{5/3}},$$

(190.5)

$$\Delta a_\tau = \frac{4\pi}{f^{4/3}}.$$

(190.6)

## 4 Refinement 1: Accuracy of the Koide formula

### Affected Documents

Doc. 116 (abstract, line 14):  $\Delta Q < 0.00003 \%$

Doc. 158 (abstract): “experimentally confirmed to 0.001 %”

### Correct Interpretation

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \quad (190.7)$$

is experimentally confirmed to better than 0.001 % (PDG values 2022). The figure  $< 0.00003 \%$  in Doc. 116 refers to the internal consistency of the T0 formulas, not the experimental measurement uncertainty. Both statements are substantively correct but refer to different quantities. In external communications, 0.001 % is the appropriate figure.

### Note

The individual T0 masses agree with experiment to within  $\sim 1 \%$ ; the high precision of  $Q = 2/3$  arises from the specific algebraic structure of the Koide combination, not from individual mass accuracy.

## 5 Refinement 2: $\hbar$ from the Higgs field - derivation chain

### Context

Planck’s constant  $\hbar$  is derived differently in various documents. The complete, consistent derivation chain is fixed here.

Binding Derivation Chain

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \longrightarrow L_0 = \xi \cdot \ell_P \longrightarrow \hbar \sim \sqrt{\xi} \cdot \ell_P \cdot c \quad (190.8)$$

The time-mass binding condition provides the physical justification:

$$T(x,t) \cdot m(x,t) = 1 \implies T = \frac{\hbar}{m c^2} \implies E = \hbar \omega. \quad (190.9)$$

$h = 2\pi\hbar$  is therefore *not* a free constant but a geometric consequence of  $\xi$  and  $\ell_P$ . The numerical SI value  $h \approx 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$  is the calibration of this relation onto the SI base units.

6 Refinement 3: Two  $\xi$  parameters - two distinct spaces

Context

In Documents 173–176 (quantum dots, spin qubits, qubit state spaces, Shor algorithm), two numerically different  $\xi$  values appear. Earlier documents did not separate these consistently, which can cause confusion.

Binding Distinction

| Parameter                           | Value                                              | Domain                       | Meaning                                                         |
|-------------------------------------|----------------------------------------------------|------------------------------|-----------------------------------------------------------------|
| $\xi_0$<br>(geometric)              | $\frac{4}{30000}$<br>$\approx 1.33 \times 10^{-4}$ | 3D geometric space           | Length hierarchy of the torus; valid in physical position space |
| $\xi_{\text{Higgs}}$<br>(algebraic) | $\approx 1.04 \times 10^{-5}$                      | Number / quantum state space | Algebraic field corrections; Higgs sector; de-coherence rates   |

Justification

$\xi_0 = 4/30000$  follows from the torus geometry of physical space (geometric derivation, independent of SI measure-

ments).  $\xi_{\text{Higgs}}$  follows from the algebraic Higgs field structure and appears in coupling formulas between quantum states (e.g. Bell correlations,  $T_2$  hierarchy).

**The two  $\xi$  values are not a contradiction**; they describe complementary aspects: geometry of space ( $\xi_0$ ) vs. algebra of state space ( $\xi_{\text{Higgs}}$ ). External communications must always specify which parameter is meant.

## 7 Refinement 4: $L_0$ and the Schwarzschild interpretation

### Affected Documents

Earlier versions of Doc. 180–182 (Lagrangian derivation, torus geometry, maximum cosmological scale).

### Erroneous Approach

Earlier versions interpreted  $L_0 = \xi_0 \cdot \ell_P$  as the Schwarzschild radius of a fundamental minimal object. This interpretation was explicitly discarded in Doc. 182 (2026 revision).

### Correct Interpretation

$L_0$  is exclusively the **geometric fundamental length** of the FFGFT torus:

$$L_0 = \xi_0 \cdot \ell_P \approx 2.15 \times 10^{-39} \text{ m.} \quad (190.10)$$

It defines the lowest scale stage of the  $\xi$  hierarchy and has *no* Schwarzschild interpretation. The cosmological Hubble radius  $R_H$  is the system-specific upper limit of the cosmological sector – not a universal maximum scale. Each scale has its own system-specific upper limit.

### Note: Schwarzschild as Consistency Check

The Schwarzschild connection remains valid as an **internal consistency check**: an object of mass  $M_0 = \xi_0 \cdot m_P/2$  would have a Schwarzschild radius exactly equal to  $L_0$  –



with no free parameter. If  $M_0$  matches a scale stage of the  $\xi$  hierarchy, that is a non-trivial self-consistency test. The connection is a *probe, not a derivation*.

## 8 Refinement 5: Scope and reach of the mass formulas

### Context

Several FFGFT documents describe mass derivations for leptons and other particles. The scope of these derivations must be communicated clearly.

### Binding Formulation

The FFGFT mass formulas (e.g.  $m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v$ ) yield values that agree with experiment to  $\sim 1\%$ . This is **not a precision calculation** but a structural proof:

- The mass formulas *follow geometrically* from  $\xi$  and  $v$  – they are not fitted.
- The remaining  $\sim 1\%$  deviations reflect measurement uncertainties and approximations, not errors in the structure.
- Numerical accuracy beyond the Standard Model is *not* claimed.

To be avoided in external texts: “FFGFT calculates the masses exactly.” The correct statement is: “FFGFT *derives* the mass structure from a single parameter  $\xi$  and reproduces the orders of magnitude to  $\sim 1\%$ .”

## 9 Refinement 6: Koide formula only with original values

### Binding Rule

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \quad (190.11)$$

must be evaluated only with experimentally measured masses (PDG values).

**Not permitted:** substituting the symbolic FFGFT terms  $r_\ell \cdot \xi^{p_\ell} \cdot v$  directly into the formula. The  $\sqrt{m_i}$  operation generates cross-terms with mixed  $\xi$  powers that are foreign to the torus structure space. Any such substitution yields  $Q \approx 0.6677 \neq 2/3$  and is not a contradiction to the theory but a category error.

**Permitted:** evaluating FFGFT terms numerically (e.g.  $m_\tau = 1783 \text{ MeV}$ ) and inserting that result into the Koide formula like a measured value.

**Mathematical background:** The  $r_\ell$  terms are geometric invariants of their torus modes. The set of admissible pairs  $(r_i, p_i)$  is not closed under multiplication:  $r_e \cdot r_\mu = 64/15$  does not belong to any torus mode. This is the Non-Closure Theorem (Doc. 193). The general analysis of composition limits is found in Doc. 192.

## 10 Refinement 7: Terminology “renormalization group” and scale structure

### Affected Documents

- Doc. 005 (T0 tm-extension): “RG flow”, “renormalization group factor”, “renormalization group invariance”, subsection “renormalization group treatment”.
- Doc. 008 ( $\xi$  and  $e$ ): subsection “modified renormalization group equations”.
- Doc. 019 (T0 Lagrangian, earlier version): subsection “renormalization group equations”.
- Doc. 086 (document overview): “renormalization group as flow in parameter space”.
- Doc. 189 (torus derivation): “cumulative renormalization group running”.

- Doc. 202 (FFGFT comprehensive field theory), §18: section title “The renormalization group”.

### Previous Formulation

In the documents listed, the  $\xi_0$ -modified scale running of the coupling

$$\frac{d\alpha}{d \ln \mu} = \beta_0 \alpha^2 \left( 1 + \xi_0 \ln \frac{\mu}{E_0} \right) \quad (\text{old - imprecise})$$

and the corresponding scale dependence are designated as the *renormalization group*, sometimes adjacent to statements explicitly characterizing  $\xi_0$  as a topologically fixed geometric constant.

### Refined Formulation

In FFGFT, the scale dependence of the coupling is *not* a result of a renormalization procedure but a direct consequence of the recursion  $\mathcal{R}$  (Doc. 203) on the fractal torus geometry. The correct designation is therefore:

|                                                                                         |          |
|-----------------------------------------------------------------------------------------|----------|
| Scale structure from recursion<br>$\equiv$ recursive $\alpha$ running via $\mathcal{R}$ | (190.12) |
|-----------------------------------------------------------------------------------------|----------|

Specifically:

- The term  $\xi_0 \ln(\mu/E_0)$  arises from the iterated application of the recursion  $\mathcal{R}$  to the mode structure, not from a counter-term subtraction.
- $\xi_0$  is *not* a fixed point of a  $\beta$  function but a topologically fixed constant ( $\xi_0 = 4/30000$ , Doc. 180).
- Stability, scale flow, and UV finiteness are three aspects of *the same* recursive structure – not separate mechanisms.

Language Convention

| Avoid                                         | Use                                                                 |
|-----------------------------------------------|---------------------------------------------------------------------|
| Renormalization group (implies QFT procedure) | Scale structure from recursion                                      |
| RG flow / RG running                          | Recursive $\alpha$ running; $\mathcal{R}$ -induced scale correction |
| Renormalization group equations               | Scale equations from $\mathcal{R}$                                  |
| Renormalization group invariance              | Recursive scale invariance                                          |
| Renormalization group fixed point             | Recursion fixed point: $\mathcal{R}(\Phi_*) = \Phi_*$               |

Justification

The designation “renormalization group” imports a tool of standard quantum field theory based on a fundamentally different principle: there UV divergences are absorbed by a procedure, and parameters *run* under a formal scale transformation. In FFGFT neither the divergence problem (cf. Doc. 159, “without artificial renormalization”) nor a free scale parameter to be absorbed exists. The scale dependence is a *structural property* of the recursion, not a subsequent adjustment step.

The FFGFT-native view is already correctly formulated in Doc. 159 (“renormalization as symptom, not as solution”) and Doc. 189 (“geometrically determined, not by renormalization”); this clarification serves to unify the terminology across the older Lagrangian and  $\xi$ -derivation series (Doc. 005, 008, 019) and across Doc. 202.

The underlying mechanism is developed in Doc. 203 (recursion operator  $\mathcal{R}$ ) and Doc. 204 (fractal boundary condition).

11 Refinement 8: Terminology “fractal renormalization”

Affected Documents

Doc. 005, 008, 012, 014 (with 19 occurrences the most frequent source), 041, 060, 133, 141, 159, 181.

Previous Formulation

“Fractal renormalization” is used as an FFGFT-internal term for the geometric correction factors in the conversion between natural and SI units, and for scale-dependent corrections on the fractal torus.

Refined Formulation

Since the word “renormalization” – even in compound form – carries the QFT connotation of a subtraction or adjustment procedure, the term is replaced by context-specific alternatives:

| Context                                                                     | Binding designation                               |
|-----------------------------------------------------------------------------|---------------------------------------------------|
| Unit conversion<br>natural ↔ SI<br>(esp. Doc. 014)                          | Geometric scale adjustment                        |
| Scale-dependent corrections<br>on the fractal torus<br>(Doc. 133, 159, 181) | Fractal correction<br>(matches title of Doc. 133) |
| General scale transformation<br>by recursion                                | $\mathcal{R}$ -induced<br>scale correction        |

Justification

The proprietary term “fractal renormalization” was meant to distinguish the procedure clearly from QFT renormalization, but it adopts the central word and unavoidably with it the connotation of a *correction procedure*. In FFGFT this is not a procedure but the direct numerical manifestation of the fractal geometry. The proposed alternatives avoid this misunderstanding without requiring new vocabulary – both phrases are already estab-

lished in the corpus (Doc. 133 “fractal correction”; “geometric scale adjustment” in the same semantic field as Doc. 014).

## 12 Refinement 9: Column “Symmetry” and the term “color charge” in the fermion tables

### Affected Documents

- Doc. 006 (T0 particle masses), “universal fermion table”, column “Symmetry”.
- Doc. 046 (Particle Masses, English parallel version), same table, column “Color”.
- Doc. 042 ( $\xi$  parameter and particles), table “spatial field patterns”, entry “Quarks: multi-node bound clusters, confined, color charge”.
- Doc. 005 (T0 tm-extension), Method 2 “extended fractal method with QCD integration”: use of  $N_c = 3$ ,  $\alpha_s$ ,  $\Lambda_{\text{QCD}}$  as external inputs.

### Previous Formulation

In the fermion tables of Doc. 006/046 each row carries a column “Symmetry” with the following entries:

| Particle                        | Entry under “Symmetry” |
|---------------------------------|------------------------|
| Electron, Muon, Tau             | Lepton number          |
| $\nu_e, \nu_\mu, \nu_\tau$      | Double- $\xi$          |
| Up, Charm, Strange, Top, Bottom | Color                  |
| Down                            | Color + Isospin        |

This mixture combines four different concepts in one column: an FFGFT scaling class (“Double- $\xi$ ”), an SM conserved quantity (“lepton number”), an SM gauge symmetry (“color”), and an SM quantum number (“isospin”). In addition, the entry “color charge” in Doc. 042 and  $N_c = 3$  in Doc. 005 suggest that the SM color symmetry  $SU(3)_C$  is an independent component of FFGFT.

Refined Formulation

In FFGFT neither the gauge group  $SU(3)_C$  nor a color quantum number exists. Quark masses follow, just like lepton masses, exclusively from the Yukawa resonance formula

$$m_f = r_f \cdot \xi^{p_f} \cdot v \tag{190.13}$$

with  $r_f$  a rational prefactor and  $p_f$  a rational exponent (step size  $\Delta p = 1/3$ ). This is verified numerically by the script `calc-resonanz-leiter.py`: all six quarks hit their PDG masses to  $\sim 0.3\text{--}3.4\%$  *without* use of  $N_c$ ,  $\alpha_s$ ,  $\Lambda_{\text{QCD}}$ , or any color quantum number.

Corrected column structure of the fermion table

The old column “Symmetry” is replaced by two clearly separated columns:

| Particle                   | FFGFT class    | scaling | SM correspondence (informational) |
|----------------------------|----------------|---------|-----------------------------------|
| Electron, Muon, Tau        | Single- $\xi$  |         | Lepton number                     |
| $\nu_e, \nu_\mu, \nu_\tau$ | Double- $\xi$  |         | Lepton number, lepton mixing      |
| Up, Charm, Top             | Cluster- $\xi$ |         | Up-type, $T_3 = +1/2$             |
| Down, Strange, Bottom      | Cluster- $\xi$ |         | Down-type, $T_3 = -1/2$           |

Meaning of the classes:

- *Single- $\xi$* :  $m = r \xi^p v$  with  $p \in \{2/3, 1, 3/2\}$ .
- *Double- $\xi$* :  $m_\nu = r_\nu \xi^2 m_e$  (additional  $\xi$  suppression relative to the charged leptons).
- *Cluster- $\xi$* :  $m = r \xi^p v$  with the same exponents  $p \in \{-1/3, 1/2, 2/3, 1, 3/2\}$  as for the leptons, but with multi-node binding in the sense of Doc. 049 (cubic self-interaction  $\lambda_s (\delta m_q)^3$ ).

Language convention for quarks

| Avoid                                                 | Use                                                                     |
|-------------------------------------------------------|-------------------------------------------------------------------------|
| “color charge”<br>(implies $SU(3)_C$ quantum number)  | Multi-node binding                                                      |
| “three colors”<br>red/green/blue”<br>(SM gauge index) | Triple-node cluster<br>(the triplet is geometric, not an $SU(3)$ index) |
| “ $N_c = 3$ colors”                                   | $N_{\text{cluster}} = 3$<br>(geometric node count in the baryon)        |
| “color confinement”                                   | Cluster binding<br>(cubic self-interaction)                             |

Status of Doc. 005, Method 2

Doc. 005 contains two calculation methods:

- *Method 1: Direct geometric method.* Yukawa form according to Eq. (190.13); parameter-free; accuracy  $\sim 1\%$  for all fermions. **This is the FFGFT method.**
- *Method 2: Extended fractal method with QCD integration.* Uses  $N_c$ ,  $\alpha_s$ ,  $\Lambda_{\text{QCD}}$ , lattice-QCD calibration and ML. **This is a phenomenological bridge construction** for connecting to the lattice-QCD literature, not part of the parameter-free FFGFT.

In external communication Method 1 is to be cited. Method 2 may be used if explicitly labelled as “bridge to lattice QCD” or “ML-calibrated extension”.

Justification

The designations “color”, “color charge”, “ $N_c$ ” import a concept from standard quantum field theory ( $SU(3)_C$  as gauge symmetry, three colors, eight gluons, color confinement as a non-perturbative property). Doc. 049 explicitly states that FFGFT contains none of these elements: “what we call ,color’ becomes high-energy node binding patterns” (Doc. 049, §3.7), with a single cubic interaction term  $\lambda_s (\delta m_q)^3$  instead of eight gluon fields.



The script validation (calc-resonanz-leiter.py, calc\_De.py) shows: all quark masses follow from the same Yukawa formula as the leptons. The column “Symmetry” of the fermion table, however, suggests that  $SU(3)_C$  is a distinguishing property of quarks within FFGFT – which is not the case.

The corrected column structure separates what is FFGFT-proper (*scaling class*) from informational SM correspondence (*SM correspondence*), in parallel with the separation of “recursive  $\alpha$  running” and “renormalization group” in Refinement 7.

## Updated Summary

| No. | Doc.                                             | Erroneous / unclear                                            | Correct / refined                                                         |
|-----|--------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------|
| C1  | 049 (HTML)                                       | $\xi = \lambda_h^2 v^2 / (64\pi^4 m_h^2)$                      | $\xi = \lambda_h^2 v^2 / (16\pi^3 m_h^2)$                                 |
| C2  | 116 (table)                                      | $r_\tau = 8/3$                                                 | $r_\tau = 25/9$                                                           |
| C3  | 018 (Explorer)                                   | $a_e = 4\pi / (f \cdot k)$                                     | $a_e = 2\pi^2 / (f \cdot k)$                                              |
| R1  | 116 / 158                                        | $\Delta Q < 0.00003\%$ vs. 0.001 %                             | Both correct; 0.001 % for external use                                    |
| R2  | several                                          | $\hbar$ as a postulate                                         | $\hbar$ follows from $\xi$ via $L_0 = \xi \cdot \ell_P$                   |
| R3  | 173–176                                          | A single $\xi$ value                                           | Two parameters: $\xi_0$ (geom.) and $\xi_{\text{Higgs}}$ (alg.)           |
| R4  | 180–182                                          | $L_0$ as Schwarzschild radius                                  | $L_0 =$ geom. fundamental length                                          |
| R5  | several                                          | “FFGFT calculates masses exactly”                              | Mass structure from $\xi$ ; $\sim 1\%$ agreement                          |
| R6  | 116 / 158                                        | Koide with FFGFT symbolic terms                                | Koide only with numerical values (non-closure, Doc. 193)                  |
| R7  | 005, 008, 019, 086, 189, 202                     | “Renormalization group” as FFGFT tool                          | Scale structure from recursion $\mathcal{R}$                              |
| R8  | 005, 008, 012, 014, 041, 060, 133, 141, 159, 181 | “Fractal renormalization” as proprietary term                  | Geometric scale adjustment / fractal correction                           |
| R9  | 005, 006, 042, 046                               | Column “Symmetry” with “Color”; $N_c = 3$ as external SM input | Columns “Scaling class” + “SM correspondence”; quarks via $m = r \xi^p v$ |